A Multi-Feature Quantile Regression Approach for Insulin Sensitivity Prediction

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***Abstract*—A multi-feature quantile regression method is presented in this work to forecast insulin sensitivity of critical care patients. The suggested approach delivers a 90% confidence interval for future insulin sensitivity by combining blood glucose measurements with insulin sensitivity values. Predicting insulin sensitivity distributions within 5th and 95th percentiles is achieved using neural network combined with quantile regression. Adapted model can scored narrower prediction confidence interval. However, the empirical converge within 90%, thus, improving the prediction efficacy. Comparing the results with the conventional one input feature model, the proposed method results perform narrower prediction confidence interval. These findings imply multi-feature method can improve glycemic control techniques and assist in making more exact insulin dosage which is clinically significance in critical care environments.**

Including BG data results in a complete depiction of patient metabolic dynamics, according

Keywords—Quantile Regression, Insulin Sensitivity Prediction, Multi-Feature Model, Blood Glucose, Neural Network, Intensive Care, Glycemic Control, STAR Protocol.

I. Introduction

Effective glycemic control in the Intensive Care Unit (ICU) remains a subject of debate for reducing patients morbidity [1], [2]. The Stochastic TARgeted (STAR) protocol, which utilizes intensive control insulin-nutrition-glucose (ICING) model to simulate the intricate dynamics of glucose-insulin interactions, highlights the importance of accurately estimating insulin sensitivity (SI) as a reflection of patient-specific metabolic states [3], [4]. STAR protocol used a model-based criteria relying on the ICING model, a pharmacokinetic model describing the metabolic system of the human body (see Fig. 1) [5], [6]. The patient's blood glucose response can be calculated with the ICING model for a given treatment configuration (nutrition and insulin dosage). This model is used especially in the STAR protocol [7]. The parameters of this model are considered constant by the protocol, except for SI [8], [9], allowing changes in the patient's condition to be described by altering the SI parameter. Traditionally, SI prediction has often relied on limited input features, typically focusing on the current SI value or a single preceding BG measurement, which is a crucial parameter in clinical settings. That may not fully capture the complex temporal interplay inherent in critical care scenarios [8-10].

Recent advances in machine learning, particularly neural network-based quantile regression, have demonstrated promising results in modeling such multifaceted physiological processes. This work extends methods by incorporating a multi-feature approach that leverages historical BG and SI measurements to predict future SI values [13], [14]. This enriched feature set allows for a more comprehensive depiction of a patient's metabolic status, addressing the uncertainty and variability present in clinical data [11].

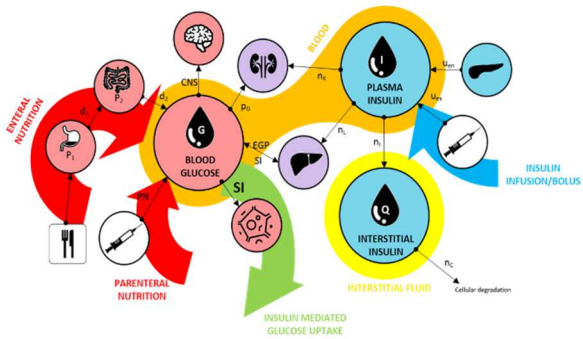

Fig. 1. Sketch representation of the ICING pharmacokinetic model. The colors represent the different compartments.

By applying quantile regression techniques, our model directly estimates the 5th and 95th percentiles of the future SI distribution, thereby providing a robust 90% confidence interval for predictions [13], [15]. This approach not only enhances the precision of the SI forecast but also offers valuable insights into the underlying patient dynamics,

facilitating safer and more effective glycemic control. In-silico validation using a large clinical dataset confirms that our multi-feature model outperforms traditional single-feature methods, marking a significant step forward in personalized ICU management [13], [14].

This study presents the development, implementation, and evaluation of our multi-feature quantile regression model for SI prediction, demonstrating its potential to refine clinical decision-making and optimize patient care in critical settings..

II. METHODS

A. Clinical Data Set

Along 12 years, clinical data was collected from patients under STAR treatment at CHU de Liege (Belgium), Christchurch Hospital (New Zealand), and Kalman Pandy Hospital (Hungary). Patients stay at ICU least 12 hours were included. The total recruited patients were 2,349, 20% of patients left back for future validation. SI values was derived from patient BG records getting 119780 SI data points.

Fig. 2 shows join distribution between SI(t) and SI (t + 1) along with probability distribution function (PDF) of SI values. The PDF curves is right skewed, meaning the majority of data located in the relatively small SI values domain. 80% of the patient's data randomly selected for purpose of training neural network, remanning data used in order to validate the proposed prediction model.

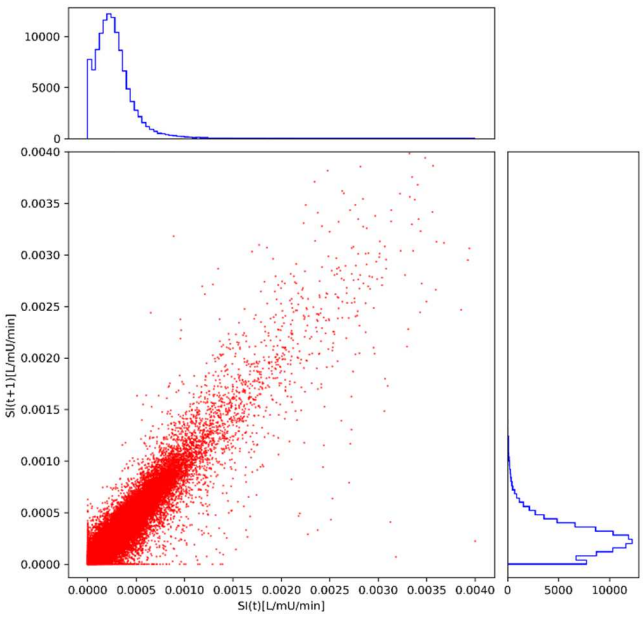

Fig. 2. Joint Distribution of SI and SI(t+1) (SIM1, SIP2 ≤ 0.0025).

B. Quantile Regression

Quantile regression is a powerful statistical tool that directly estimates the conditional quantiles of a target variable rather than its mean and variance using the benefits of pinball loss function. This approach allows us to identify a robust 90% confidence interval by estimating 5th and 95th percentiles borders of the SI distribution [16].

By representing y as the true SI(t+1) and $\hat{y}$ the predicted quantile at the quantile level τ (where $0 < \tau < 1$), the predicted error can be calculated using the pinball function:

$$L_\tau(y,\hat{y}) = \begin{cases} \tau(y-\hat{y}) & if\ y \geq \hat{y} \\ (1-\tau)(y-\hat{y}) & if\ y \leq \hat{y} \end{cases} \quad (1)$$

Graphical representation of pin ball loss function shown in Fig. 3, interpret the quartile error and the penalty shift, is asymmetric for different quartiles [17].

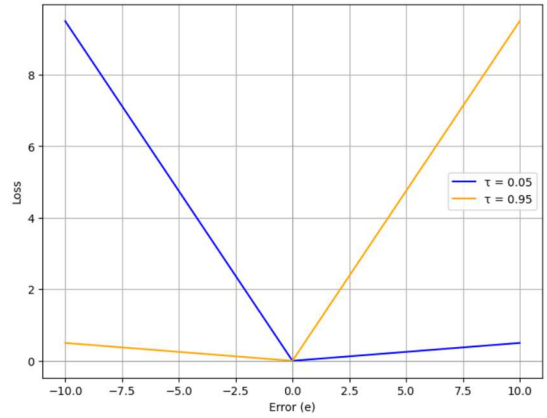

Fig. 3. The loss function of pinball for two quantiles τ=0.05 and τ=0.95.

TensorFlow within Python work space was employed to implement shallow neural network combined with quantile regression for SI prediction approach, the model trained for 100 epochs using L1 and L2 regularization to prevent over fitting. Neural network configuration shown in the Table 1, SI(t) and current BG used as input features, while 5 and 95 percentile of next SI time horizon as target outputs. The network has been optimized by minimizing the combined pinball loss for the lower and upper quantiles:

$$L_{total} = L_{0.05}(y,\hat{y}_{0.05}) + L_{0.95}(y,\hat{y}_{0.95}) \quad (2)$$

where, L_{total} is total loss, $L_{0.05}$ and $L_{0.95}$ is the 5 and 95 loss quantile, respectively. y is the true SI(t+1) value and $\hat{y}$ is the predicted next SI values.

TABLE I. The topology is used to train the network.

Layers	Number of neurons	Activation Function
Input	2	
Hidden	4	Selu
Output	2	linear

The anticipated 5 and 95 percentiles of this model create a 90% confidence interval for the following SI value. This multi-feature method increases the general accuracy and resilience of glycemic control techniques by efficiently capturing the temporal dynamics of insulin sensitivity and the interaction with blood glucose levels.

III. PERFORMANCE METRICS

Three case-specific metrics have been employed to evaluate the SI prediction, assessing both of the reliability and precision of the predicted 90% confidence intervals. These metrics are the success rate (SR), interval ratio (IR), and I-score.

A. Success Rate

It quantifies the fraction of observed SI(t+1) values within the predicted confidence interval. N_{in} represents the collected data sets whose true SI values lie between the predicted lower and upper bounds, N is the total observations data [16], [18]:

SR near 0.9 is preferable

$$m_{SR} = \frac{N_{in}}{N} \tag{3}$$

B. Interval Ratio

The IR evaluates the relative width of the predicted confidence interval compared to a interval width provided from kernel predictor in reference model STAR. Let r_i denote the width of the STAR interval for the i-th data point, and ($h_i - l_i$) be the width of the predicted interval. The IR is calculated as [16], [18]:

$$m_{IR} = \frac{1}{N}\sum_{i=0}^{N}\frac{h_i - l_i}{\tau_i} \tag{4}$$

An IR value below 1 suggests that the model yields a narrower and more precise interval than the reference model.

C. I-Score

The I-Score combines both the SR and the IR into a single metric, balancing calibration and precision. It is defined by [16], [18]:

$$m_{IS} = \frac{icdf(0.5 + \frac{m_{SR}}{2})}{icdf(0.95).m_{IR}} \tag{5}$$

where, icdf() represents the inverse cumulative distribution function of standard normal distribution data. I-score of 1 indicates an ideal predictor, while a value greater than 1 points to improved performance relative to the baseline.

IV. Results

A 20% hold-out test set was used to properly analyze the multi-feature prediction model, which combines BG and SI values. Leveraging historical data from several time steps, the model is meant to provide a 90% confidence interval for the future SI value SI(t+1). This method uses neural network-based quantile regression to reflect patient metabolic response natural variability. Three main criteria IR, SR, and I-Score allow one to evaluate the model.

Calculated to be 0.662, the interval ratio, that is, the ratio of the width of the model-generated confidence interval relative to a standard interval was This indicates more accuracy in the forecasts as the model-generated confidence range is only 66.2% as broad as the conventional interval. A smaller gap indicates that the model can more precisely and deliberately project the future SI value.

Apart from accuracy, calibration is quite important for evaluating forecast dependability. Found to be 0.889, SR defined as the percentage of actual SI(t+1) readings inside the expected 90% confidence interval was This number indicates that the interval of the model effectively captures 88.9% of the observed data points, thus verifying strong calibration even if it is somewhat below the goal value of 0.90.

Moreover, the I-Score—which aggregates accuracy and calibration by means of an inverse cumulative distribution function between the reference and forecast intervals—was found to be 1.463. An I-score higher than one shows a good balance between limited interval and strong coverage of the real SI values. This measure supports the general efficiency of the multi-feature technique in precisely forecasting SI values in the future.

Fig. 4 provides a detailed scatter plot of SI(t) versus SI(t+1). In this visualization, the observed SI(t+1) values are marked by red dots, while the model-generated 5th and 95th percentile lines are displayed in blue and orange, respectively. The figure clearly illustrates that the majority of the data points fall within the predicted 90% confidence interval, visually corroborating the SR of 88.9%.

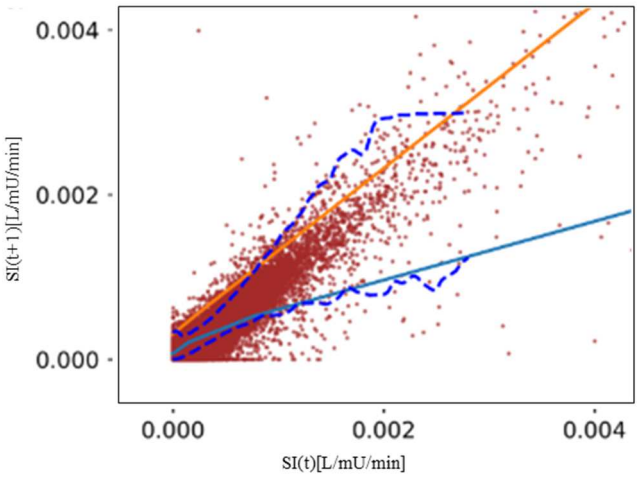

Fig. 4. The scattering plot of SI(t) versus SI(t+1).

Fig. 5 complements this analysis by presenting the training and validation loss curves over multiple epochs. Both curves exhibit a steady decline and converge smoothly, with minimal divergence between them. This behaviour reduces the possibility of overfitting by indicating that the model has successfully learnt the underlying patterns in the multi-feature data and is generalizing well to unknown data.

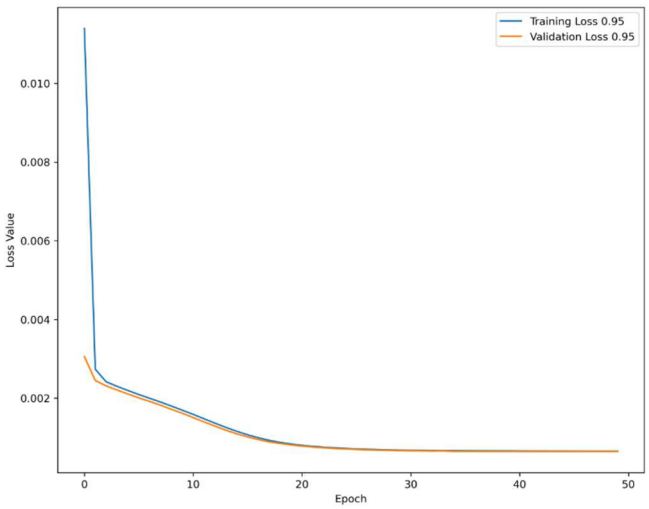

Fig. 5. training and validation loss curves over multiple epochs.

These results indicate that incorporating both BG and SI features into the prediction model significantly improves the precision and reliability of future SI estimations. The model's narrow confidence intervals (IR = 0.662), capture rate (SR = 0.889), and balanced performance (IS = 1.463) underscore its potential utility in clinical settings. Accurate prediction of SI is essential for guiding insulin dosing in intensive care, and the promising performance metrics of this multi-feature

approach highlight its potential to enhance glycemic control protocols and improve patient outcomes.

The comparison between the current multi-feature prediction model and previous SI prediction approaches is reported in the literature. The comparison is based on three primary performance metrics: SR, IR, and I-Score. These metrics collectively assess both the calibration and precision of the predicted 90% confidence intervals for future SI values.

TABLE II Comparison between current work and other related work shows the three key performance parameters.

Reference	Method	S.R.	I.R.	I-Score
[16]	1D Prediction	0.880	0.680	1.490
	2D Prediction (Polynomial)	0.900	0.663	1.517
	3D Prediction (Polynomial)	0.908	0.701	1.477
Current	Multi-feature Prediction	0.889	0.662	1.463

The comparative analysis highlights that while the 3D polynomial model exhibits the highest success rate, the multi-feature approach offers a favorable balance between precision and reliability. Integrating BG measurements with SI data results in narrow confidence intervals and robust calibration, underscoring the potential of multi-feature models to support more informed clinical decision-making in intensive care settings.

V. Discussion

Promising performance in estimating future SI values has come from the multi-feature quantile regression model, which combines BG and SI observations. On a 20% hold-out test set, the model obtained an SR of 0.889, an IR of 0.662, and an I-Score of 1.563, implying that its 90% confidence intervals are tight and consistently calibrated.

These results have rather significant clinical consequences. Dosing insulin in critical care environments depends on accurate SI prediction; in these cases, the unpredictability in patient response may cause dangerously fluctuating BG levels. By reducing the risk effect of hypoglycemia and hyperglycemic events, the multi-feature method improves SI prediction by using the dynamic interaction between BG and SI and helps to safer glycemic management.

Despite these promising results, certain limitations remain. The success rate, although high, is slightly below the ideal target of 0.90, and the I-Score, while favorable, indicates room for further optimization. Future research should focus on exploring additional physiological parameters, experimenting with alternative neural network architectures, and fine-tuning quantile regression parameters to further enhance both calibration and precision.

VI. Conclusion

The work shows that using a multi-feature quantile regression model that combines blood glucose and insulin sensitivity measurements predicts future values of SI more accurately than the quantile regression model using only SI values. The model presents competitive performance to conventional prediction algorithms with a Success Rate of 0.889, an Interval Ratio of 0.662, and an I-Score of 1.493. Including BG data improves the model by capturing thorough temporal patterns underlying in-patient metabolism, providing narrower and more accurate confidence intervals. Although the results are encouraging, more research, including network design optimization and potential inclusion of further physiological parameters into the prediction model, are required before initializing prospective clinical trials. The results, in general, highlight the possibilities of the multi-feature approach in improving glycemic control recommendations to reach better patient outcomes in critical care settings.

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